

Influence of ChatGPT on Wikipedia - Group 22

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1 INTRODUCTION

Wikipedia is a medium for global knowledge, collaboration and information distribution that strongly influences education and research. Understanding the trends in user behaviour, edit patterns, and content consumption over the past 11 years (2014-2025) gives valuable insight into how much the platform usage has been changing over the years and how external factors have been influencing those changes. In this paper, we will be taking a closer look at the impact generative Artificial intelligence (AI) has on Wikipedia and its user traffic in correlation with a major world event, the war in Ukraine. The reason we chose the war in Ukraine is because it is a major world event that has been going on since 2014 and is ongoing at the moment this paper is written. There are numerous articles online especially on Wikipedia about the war, with a lot of newly created articles and multiple revisions. The context of the war in Ukraine also makes it a controversial event where certain parties might have an interest in influencing the opinion of the reader through renowned informative platforms such as Wikipedia.

The main proposed **research question** is:

"What are the trends in Wikipedia usage for the past 15 years regarding new user accounts, article view count, and number of edits?"

The main question will be answered using these three **subresearch questions**:

- 1) *What are the aspects that indicate the use of generative AI?*
- 2) *How are the trends affected by the occurrences of generative AI?*
- 3) *How are the trends affected by major world events?*

By analyzing the quality and authenticity of Wikipedia content revisions, we can see the impact of generative AI on such a renowned informative platform. The project also tries to identify specific revision characteristics such as verbosity, and certain text patterns made of specific words or phrases often associated with generative AI. Moreover, analyzing the user traffic on pages about significant world events (global crises, political events, and technological developments) gives away the role of Wikipedia nowadays compared to more than 10 years ago.

The following outcomes could be achieved by conducting the above-mentioned analysis:

- It reveals patterns in the average number of user registrations, article views, and edits per article.

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- Identify how generative AI tools influence editing patterns and account registrations, and shape the available information on Wikipedia. This can provide strategies to manage such content and adopt rules for quality and authenticity.
- Detect edits done by generative AI by analyzing certain characteristics (words, content length, etc.), and ensure information reliability.
- Study the traffic on Wikipedia and notice the platform's importance as a real-time, collaborative resource during times, or about topics of major interest.

These outcomes could guide Wikipedia administrators in creating guidelines to ensure accuracy and reliability on the platform.

2 RELATED WORK

Other studies around Wikipedia in combination with ChatGPT exist but mainly focus on the effects AI has on Wikipedia's engagement. We were not able to find anything similar to what we are doing, perhaps because it is rather focused on figuring out if there are AI influences over controversial Wikipedia Articles, such as the war in Ukraine.

Our project idea can be considered generic but the topic we chose, in combination with the results we are aiming to obtain from the data, is quite niche. The gap we are aiming to fill is determining if AI is used as a way to manipulate Wikipedia articles related to the war in Ukraine.

3 METHOD

3.1 Dataset

In general, all the data has been obtained from *Wikimedia* [1] and *MediaWiki* [2] APIs, which are two of the many components that Wikipedia has behind to enable access to its data. An explanation of the criteria for choosing the data can be found below.

3.1.1 Articles. We have chosen to focus on the war in Ukraine as our main topic because it is a recent and ongoing event. The full scale of the event would be from 2014 till the moment of writing this paper. This means that it still receives attention from users in terms of reading and editing existing content, as well as creating new content. This also ensures that related articles will likely have undergone numerous recent edits and revisions, allowing us to analyze the impact of the pre-ChatGPT and post-ChatGPT era effectively.

3.1.2 Collecting the data set. Every article on Wikipedia is tagged with multiple "Categories", and they are accessible for us to use. Therefore, we used these categories as the basis for obtaining a large number of relevant articles.

The detailed method for selecting the articles is as follows:

- (1) Manually identify popular Wikipedia pages that lead directly to the main subject; the war in Ukraine.

- (2) Retrieve the *Categories* these popular pages are tagged with.
- (3) Retrieve all the pages included in the categories obtained in the previous point.
- (4) Extract the categories from the newly identified articles.
- (5) Retrieve all the pages included in the newly identified categories, effectively going one layer deeper in terms of categories.

3.1.3 Article Revisions. The revisions are the main working dataset

- (1) Revisions
Needed for getting the content as well as other relevant information such as the byte size of the change, the user who made the change, etc.
- (2) Revisions Content
Comparing sequential revisions gives us the actual content of the change, allowing us to identify patterns across time.

Attribute	Value
Type	Type of revision
Line Number	Where the change appears
Text	Changed text
Offset	Location of the line in bytes from the beginning of the page
Highlight Ranges	Range on the page
From ID	Id of Previous revision
To ID	Id of this revision
Title	Title
Date	Timestamp

Table 1. Data Metrics

3.1.4 *Analytics*. A part of the analysis of the project analytics was used as a supplementary dataset to gather more insights Those being:

- (1) New users over time
- (2) Page views over time
- (3) Edits over time

When it comes to data size, the uncompressed values are as follows:

Metric	Value
Number of articles	67,297
Data size of articles' content	1.48 GB
Data size of revisions	1.7 GB*

Table 2. Data Metrics

The above-mentioned data is stored in .json formatted files on the HDFS.

3.1.5 *Challenges in obtaining data*. The process of fetching data about the revisions of each article required a large number of API calls, which will result in a "429: Too Many Requests" error code roughly every 10 minutes. Hence, we decided to design the Spark code as it would be for big data even though the data is not that big to overcome this obstacle.

Since we were only able to obtain through API calls roughly 100 MB of revisions from 4,000 articles (a few hundred thousand entries), our estimations say that if this process could have been automated in a way that prevented the 429 codes from happening, we could have been looking at roughly 1.7 GB* of revision data and roughly 2.5 million entries. These are numbers based on the fact that we have a total of 70,000 articles, with approximately 100 MB, and 150,000 entries for every 4,000 articles.

3.1.6 *Uncommon word count*. Data was taken from the HDFS directory containing revision histories of multiple documents, structured as JSON files. The text from each revision was processed by aggregating all revisions grouped by their unique identifiers (to_id) and timestamps. Before analysis, the text was cleaned to remove non-alphabetic characters, ensuring a uniform comparison of word usage. For each revision, the total number of words not present in the common ChatGPT word list was computed. The resulting data consisted of the revision identifier, timestamp, and the count of uncommon words. The uncommon word counts were aggregated by year and month to identify long-term trends. Only data from 2020 onwards was included in the analysis to ensure relevance to the timeline of ChatGPT's influence. Records were grouped by their year and month to produce a time series for trend analysis.

3.2 Metrics

To answer the question regarding the impact of Large Language Models on Wikipedia, 3 potential metrics were discovered to do so:

3.2.1 *Length of revision*. ChatGPT's answers are typically longer than those of humans [3]. Therefore, analyzing patterns in the character length of each revision could provide useful insights.

3.2.2 *Use of common words used by ChatGPT*. ChatGPT's vocabulary differs from that of a regular human, especially when discussing scientific and factual topics. Therefore, analyzing an increase in the usage of such terms could prove valuable. To compile a representative list of commonly used ChatGPT words, we relied on external analyses of its generated text. Specifically, these analyses identified phrases and vocabulary that ChatGPT frequently employs across various contexts, particularly in professional, academic, and technical writing [4, 5]. The resulting list served as a reference to filter and track the presence of these terms within Wikipedia revisions over time.

3.2.3 *Sentence length*. Sentences generated by LLMs are usually much more consistent in length compared to those written by humans, who may write in bursts or use very long sentences [6].

3.3 Runtimes

Check Table 3.

Metric	Runtime	Executor NR
Relative Variance	89s	6
Character Count of Revision	56s	5
Uncommon Words per Revision	7.5mins	10

Table 3. Runtime and Executor NR for Different Metrics algorithms

4 RESULTS

4.1 Analytics

To support our analysis of the effects AI has on Wikipedia, we created a series of analytics to give us more insight into some trends and how they are affected by AI. The trends we chose are about the new user accounts created on Wikipedia, the number of views, and the number of edits articles are getting. For the articles, we focus on the set relevant to the topic of the war in Ukraine.

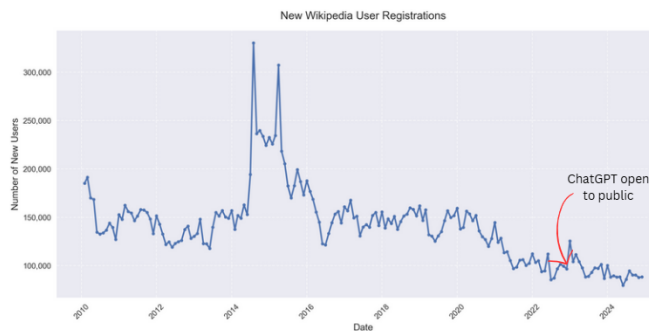


Fig. 1. New Wikipedia users 2010-2025

4.1.1 Number of views. Figure 1 is the data visualization in the form of a line chart of the number of new users on Wikipedia for the past 15 years. After analyzing the graph, two large spikes can be seen in August 2014, and April 2015 which were possibly driven by a change in Wikipedia's policy meant to incentivize users to create new accounts and obtain steward privileges (a form of moderator). Ever since, the trend line has been decreasing all the way till the current moment this paper is written, with a small spike after the release of ChatGPT to the public on November 30th, 2022.

4.1.2 Number of views. Figure 2 is the data visualization for the total number of views for the war in Ukraine topic from the past 5 years. A large spike can be seen on the 24th of February 2022, roughly 1,356,000 views, corresponding with the beginning of the Russian invasion of Ukraine. As expected, the views on the 23rd of February were much lower, roughly 140,000 views. The trend is again lowering, perhaps indicating that people were getting their information from other places.

4.1.3 Number of edits. Figure 3 is the data visualization for the total number of edits, also called revisions, for the war in Ukraine topic from the past 5 years. Two large spikes were seen in February 2022 and September 2023, marking the start of the invasion and a significant military operation by Ukraine in Crimea respectively.

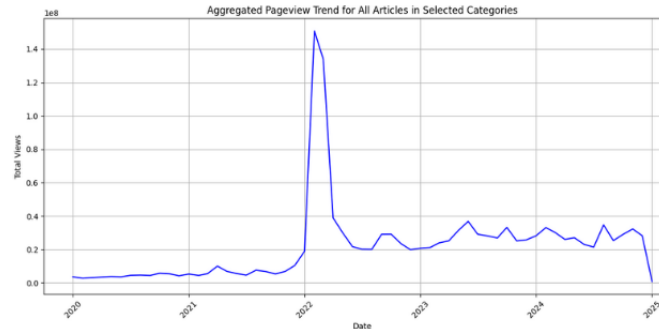


Fig. 2. Total number of views 2020-2025

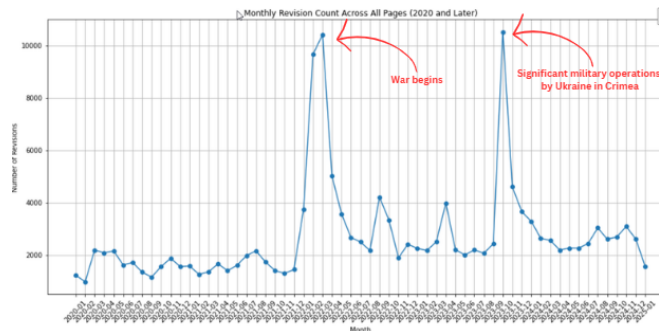


Fig. 3. Count of revisions 2020-2025

Two other smaller spikes can be noticed in August 2022, representing an escalation in Crimea and Ukraine's independence day, and in March 2023, marking a large-scale missile attack and Vladimir Putin's visit to Mariupol.

4.1.4 Uncommon Words Over Time. The analysis focused on the evolution of uncommon words across revisions over time, particularly in the context of ChatGPT's release and its influence on textual trends. The primary metric analyzed was the frequency of uncommon words that were not part of a predefined list of common ChatGPT words.

To investigate the growth in uncommon word usage, data was aggregated from 2018 onward. The frequency of uncommon words was plotted against time, with a 3-month rolling mean applied to smooth short-term fluctuations. Additionally, the release of ChatGPT (November 2022) was marked as a significant milestone in the analysis, represented by a vertical dashed green line in the plot (Figure 5). Two mean values were calculated: the mean total uncommon words before ChatGPT's release (130,521.26) and after ChatGPT's release (398,267.41), both indicated as horizontal dashed lines in the figure.

Figure 5 shows a clear increase in uncommon word usage after the introduction of ChatGPT, suggesting a potential correlation between ChatGPT's release and a rise in textual complexity or specificity. This trend was accompanied by several spikes in uncommon word usage, possibly influenced by specific events or periods of heightened textual activity.

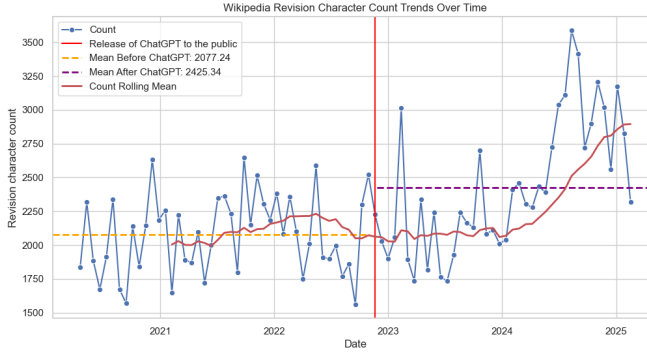


Fig. 4. Character count

To ensure the reliability of this analysis, the distribution of the data points over time was also examined. Figure 6 shows the frequency of records by date, confirming a consistent distribution across the analyzed timeline, thus ruling out data sparsity as a confounding factor.

4.2 Metrics

4.2.1 Revision Length. The first method we used to identify the presence of generative AI was through revision length since ChatGPT tends to provide long answers [6]. In Figure 4 the results show a significant increase after the release of ChatGPT, which can indicate influences on the revision content.

Process - The text is first cleaned by replacing special characters, except for letters, numbers, spaces, and common punctuation like ., , !, and ?, with spaces, ensuring that non-null values are properly formatted. A new table is then created by selecting only the columns date, text, and to_id. The number of characters in the text column is counted and stored in a new column called text_length. To facilitate data organization, a unique identifier (date_to_id) is generated by concatenating to_id and date. The table is then grouped by date_to_id, summing the total character count for each entry.

4.2.2 Uncommon word use. Another method used to determine the use of generative AI in the Wikipedia articles' content is to count the uncommon words. What we classify as uncommon words are frequently used words by generative AI such as ChatGPT [4, 5]. Examples of such words are: explore, captivate, tapestry, etc. To visualize the data we created a frequency counter for these words among all the revisions of the articles from the war in Ukraine dataset.

To visualize trends in uncommon word usage, the aggregated data was plotted as a time series. A 3-month rolling mean was applied to smooth short-term fluctuations and reveal long-term patterns. The release of ChatGPT in November 2022 was marked as a significant milestone in the timeline. Additionally, the mean total uncommon words before and after ChatGPT's release were calculated and plotted to facilitate a comparative analysis. The temporal distribution of the data was analyzed to ensure no periods of data sparsity or irregularities. A histogram of record counts by date was plotted (Figure 6), confirming consistent data coverage throughout the analysis

timeline. The data processing pipeline was implemented using PySpark for scalable distributed computation and Python libraries such as Matplotlib for visualization. The structured approach ensured reproducibility and reliability in deriving insights from the data. The results were in line with our hypothesis that there would be a growth in the use of such words which can be seen in Figure 5. There are visible spikes around and after the release of ChatGPT to the public, and to better identify patterns in the plot, we added a rolling mean and a split mean that better highlight the increase in uncommon word usage. To ensure data reliability, we created a histogram as well for the count of records across the timeline, as seen in Figure 6. The result confirms that the data is evenly distributed and therefore reliable.

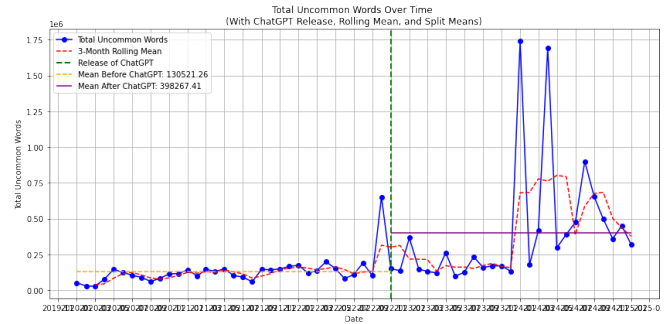


Fig. 5. Total Uncommon Words Over Time (with ChatGPT Release, Rolling Mean, and Split Means). The green dashed line represents the release of ChatGPT. The red dashed line shows the 3-month rolling mean.

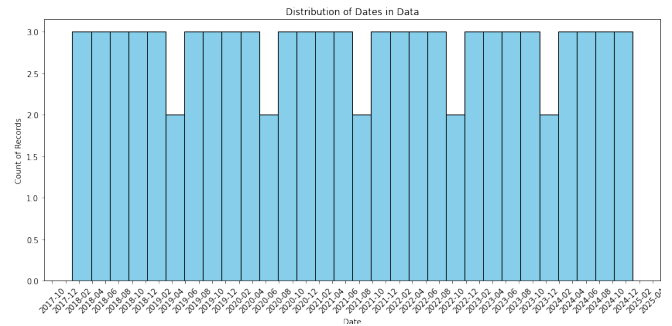


Fig. 6. Distribution of Dates in the Data. The histogram confirms consistent record distribution across the timeline, ensuring data reliability.

4.2.3 Sentence Length consistency. The final method used to determine the use of generative AI is through sentence length consistency [6]. The visualization can be found in Figures 7 and 8, and only shows a small increase in the relative variance, which would mean the revisions had more consistent sentence length in the past than currently showing no clear impact of ChatGPT. However, when looking at Figure 8, there are higher spikes in the average (red lines) than in the average word count (blue lines), which hints towards unusually long or short sentences. They can be seen in 2020, 2021, late 2023 (after ChatGPT release), and early 2025.

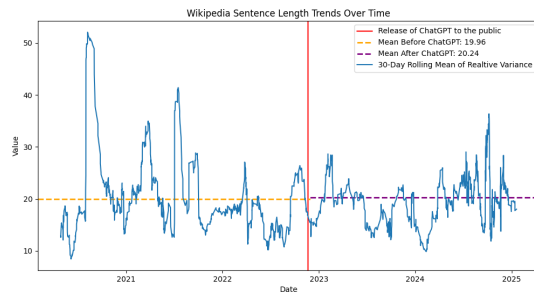


Fig. 7. Sentence length consistency

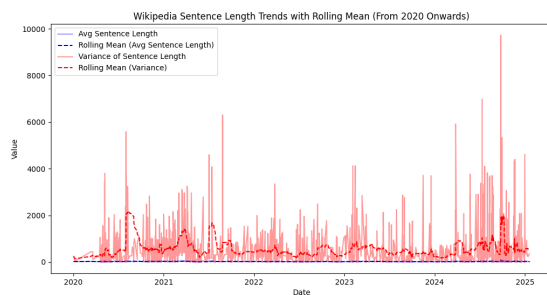


Fig. 8. Sentence length variance

Below is a short description of the method used for compiling this metric.

Process - First, the text is cleaned by removing special characters, except for letters, numbers, spaces, and common punctuation like ., ,, !, and ?, while also ensuring that no entries are left empty. Next, the cleaned text is split into sentences using ., !, or ? as dividers. A function is then applied to count the number of words in each sentence, creating a new column with these counts. To organize the data, a unique identifier (date_to_id) is generated by combining to_id and date, and unnecessary columns are dropped. The sentence word counts are then expanded so that each row represents a single sentence's word count. The average sentence length (mean word count) is computed for each date_to_id, followed by the variance of sentence lengths, which measures how much they fluctuate. These mean and variance values are then merged into a single dataset. Finally, the **relative variance**, or variance-to-mean ratio, is calculated to help analyze variations in sentence length across different entries.

4.2.4 Conclusions from metrics. As shown in Table 4, two of the three metrics indicate a considerable difference between the pre- and post-ChatGPT state of Wikipedia. However, establishing direct causality is challenging, as multiple factors may have contributed to these changes. While the data suggests a shift, other underlying influences must also be considered.

Metric	Before	After	% Change
Relative Variance	19.96	20.24	1.40 %
Character Count of Revision	2077.24	2425.34	16.77 %
Uncommon Words per Revision	0,13	0,39	205.1 %

Table 4. Comparison of Metrics Before and After with Percentage Change

5 DISCUSSIONS AND CONCLUSIONS

5.1 Discussions

Our findings can be particularly interesting to multiple categories of people such as researchers in sectors like information sciences, AI, manipulation, world conflicts, or state entities looking to analyse how information is altered through generative AI.

5.1.1 Limitations. There are multiple limitations that we identified throughout the development of the project. Some of them are a matter of knowledge and others are a matter of data. When it comes to knowledge, our main focus was on English articles, solely based on titles, with little time allocated to fact-checking the contents. This means that we cannot yet accurately also predict if the generative AI was used in a manipulative way or in a legitimate way. English articles are also just a part of what Wikipedia is, but there are also a lot of other languages, and their numbers could most probably provide interesting data, but the lack of knowledge about these languages makes it a limitation. Another limitation is caused by the data, which can be hard to collect through API requests, as also described above, because of 429 error codes, and is also not truly big data. We tried to work around these limitations, but it was too late in the project when we realised how significant these issues would become, therefore we stuck with it by making everything big data-ready. Finally, a limitation revolves around generative AI itself. There are methods to identify if the text is generated by AI but due to the nature of our project, this would require large amounts of text to be checked by an external application, which is unfeasible. Therefore, we rely on alternative methods such as sentence length and uncommon words, as described above.

5.1.2 Answer to sub-research questions. In this section, we will provide the answers to the sub-research questions in an attempt to conclude the main research question. The first question was aimed at finding what aspects can indicate the use of generative AI in Wikipedia articles. As a result, we settled on three aspects: character count of revisions, usage of uncommon words, and sentence length consistency. The answer to the second question identifies small differences in the case of sentence length but significant results in the case of character count and uncommon words after the release of ChatGPT to the public. To answer the third question, we chose a dataset of articles and revisions related to the war in Ukraine, which is a controversial and still relevant topic. This helped us to obtain a clear idea of how the trends were affected in correlation with a major world event.

5.2 Conclusion

In conclusion, the brief answer to the main research question is that the trends regarding new user accounts, article view count, and number of edits in the past 15 years were very likely to have been influenced by the appearance of generative AIs like ChatGPT. Trends such as new user accounts have been fluctuating in numbers for years, but we have noticed a constant decline for the past 5 years and article view counts and number of edits are heavily influenced by major world events. In the editing process, generative AI might play an important role, but the capacity to manipulate articles and readers’ opinions with the help of AI is something that should be investigated further.

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A APPENDIX

A.1 AI Tools

During the preparation of this work, the group used ChatGPT to aid with writing the code, clarifying ideas, and performing grammar checks. After using this tool/service, we thoroughly reviewed and edited the content as needed, taking full responsibility for the final outcome.